

HAT-P-54b

R-1642

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_181220 · created 2026-08-02T07:01:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458472.8181 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.019
Transit depth (Rp/Rs)^2	4.0 +/- 0.77 [%]
Orbital Inclination (inc)	85.67 +/- 1.5
Airmass coefficient 1 (a1)	2.091 +/- 0.056
Airmass coefficient 2 (a2)	-0.71 +/- 0.31
Scatter in the residuals of the lightcurve fit is	2.7 %
Best Comparison Star	#2 - [470, 380]
Optimal Aperture	3.51
Optimal Annulus	17.44
Transit Duration (day)	0.0255 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.019 vs 0.1572 (deviation 2.21σ)
Duration fit vs published	0.0255 d vs 0.075 d (deviation 8.81σ)
Mid-time O-C	-4.0 min
Depth SNR	10.3
Residual scatter	2.7 %

Light curve

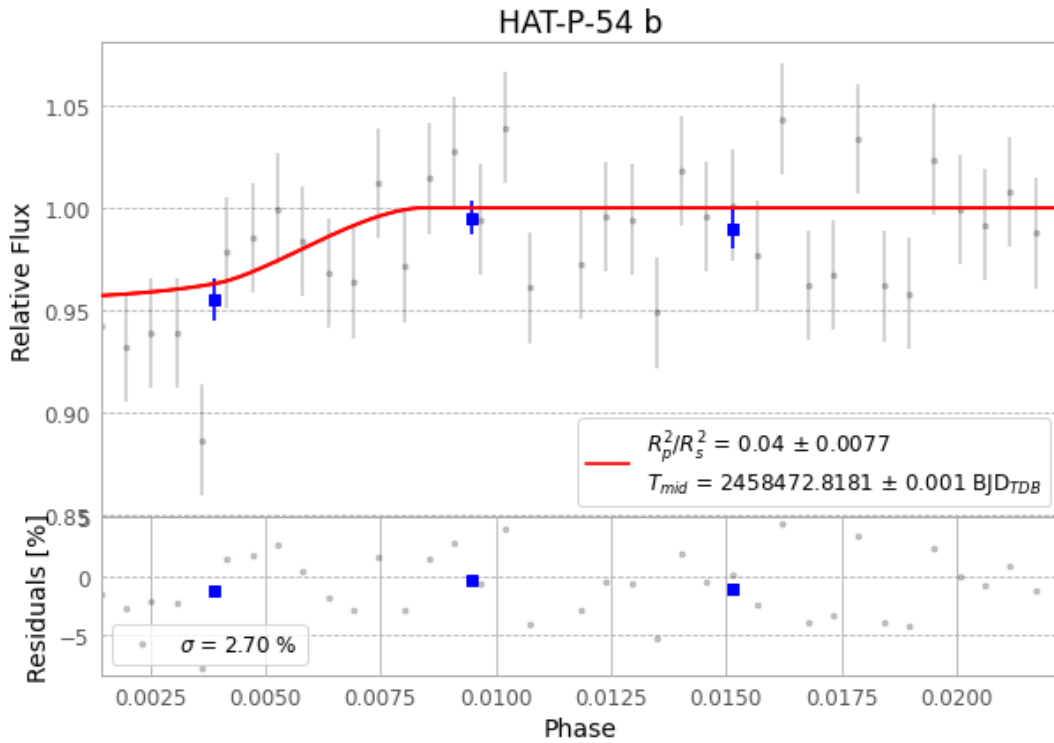


Figure 1 — FinalLightCurve_HAT-P-54 b_2018-12-20.png (SHA-256 50a399542ef8175e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3713d3aad8f56aa5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

39 FITS frames (26 MB), HATP-54181220073612.FITS → HATP-54181220093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-181220.log (99a09f641280...), FinalLightCurve_HAT-P-54 b_2018-12-20.png (50a399542ef8...), FinalLightCurve_HAT-P-54 b_2018-12-20.csv (3840cd99fe47...)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:01Z.